

Vector Algebra, Relative Velocity & Projectile Motion

1. Scalars & Vectors

Scalar: A physical quantity that has only magnitude.

Examples: Mass, Time, Temperature, Speed.

Vector: A physical quantity that has both magnitude and direction.

Examples: Force, Velocity, Acceleration, Displacement.

2. Vector Algebra – Key Formulas

- A vector is written as $\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}$

Magnitude: $|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$

Dot Product: $\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$

- Used to find the angle between two vectors.

Angle Between Vectors:

$$\cos\theta = (\mathbf{A} \cdot \mathbf{B}) / (|\mathbf{A}||\mathbf{B}|)$$

Resultant of Two Vectors:

$$\mathbf{R} = \mathbf{A} + \mathbf{B} = (A_x + B_x)\mathbf{i} + (A_y + B_y)\mathbf{j} + (A_z + B_z)\mathbf{k}$$

$$|\mathbf{R}| = \sqrt{R_x^2 + R_y^2 + R_z^2}$$

3. Relative Velocity (River–Boat Problems)

- Actual motion = boat velocity (perpendicular) + river velocity (horizontal).

Time to Cross River: $t = \text{width} / \text{boat perpendicular speed}$

Drift Downstream: $\text{Drift} = \text{river speed} \times \text{time}$

Used in problems involving motion in two dimensions.

4. Projectile Motion

Velocity Components:

$$u_x = u \cos\theta$$

$$u_y = u \sin\theta$$

Maximum Height: $H = \frac{u_y^2}{2g}$

Time of Flight: $T = \frac{2u_y}{g}$

Horizontal Range: $R = \frac{u^2 \sin 2\theta}{g}$

Motion is parabolic and vertical/horizontal motions are independent.

Work, Energy & Momentum – Detailed Physics Notes

1. Work Done by a Variable Force

Work is defined as the dot product of force and displacement.

For a variable force, the magnitude of force changes with position. In such cases, we cannot use the simple formula $W = F \times s$. Instead, we use the calculus form:

$$W = \int F(x) \, dx$$

For the force $F(x) = kx$:

$$W = \int kx \, dx = k \int x \, dx = k (x^2/2)$$

Evaluating from 0 to x :

$$W = (1/2) k x^2$$

This shows that work done by a spring-like force increases quadratically with displacement. Physically, this represents the energy stored when stretching or compressing elastic systems. **Important points:**

- Work depends on the path and variation of force.
- Area under the Force–Displacement graph gives total work.
- For $F \propto x$, the graph is a straight line; area is a triangle.

2. Impulse, Momentum & Average Force

Momentum (p) is defined as the product of mass and velocity: $p = mv$.
It represents the quantity of motion of a body.

Impulse (J) is the change in momentum:

$$J = \Delta p = p_{\text{final}} - p_{\text{initial}}$$

Impulse is also the product of average force and time duration:

$$J = F_{\text{avg}} \times \Delta t$$

Thus, we obtain:

$$F_{\text{avg}} = (\Delta p) / (\Delta t)$$

This is useful when forces act for short intervals (collisions, bullets, hitting a ball).

Key Concepts:

- Momentum is a vector quantity.
- Change in momentum occurs due to force acting over time.
- A large force acting for a small time or small force for a large time can produce the same impulse.
- Impulse is represented by the area under Force–Time graph.

Applications:

- Airbags increase time of impact → reduce force.
- Helmets reduce change in momentum impact force.
- Sports (cricket, football) rely heavily on impulse control.

3. Work–Energy Theorem & Kinetic Energy

The work–energy theorem states:

Work done by a force = Change in kinetic energy

To derive KE, start from:

$$W = \int F \, dx$$

Using Newton's law: $F = m \, dv/dt$

And $dx = v \, dt$

$$W = \int m \, (dv/dt) \, v \, dt$$

$$W = \int m \, v \, dv$$

On integration:

$$W = (1/2) m v^2 - (1/2) m u^2$$

Thus, kinetic energy is:

$$\mathbf{KE = (1/2) m v^2}$$

Important Notes:

- KE depends on square of velocity → doubling v increases KE four times.
- KE is always positive (scalar quantity).
- Net work done on a system changes its kinetic energy.
- If work done is negative (retarding force), KE decreases.

Examples in real life:

- Vehicles require more fuel at high speeds due to $KE \propto v^2$.
- Bullets, meteors, moving machines require KE calculations for safety and design.

4. Spring Energy, Retarding Force & Collisions

Elastic Potential Energy

For a spring compressed or stretched by x :

$$U = (1/2) k x^2$$

This is derived from work done against the variable spring force $F = kx$.

Retarding Forces

When a moving body is stopped (bullet in wood, vehicles braking), the force is opposite motion.

Average retarding force:

$$F_{avg} = \Delta p / \Delta t$$

Negative sign indicates opposite direction.

Momentum Conservation in Collisions

Total momentum before collision = Total momentum after collision

For perfectly inelastic collisions (bodies stick together):

$$(m_1 u_1 + m_2 u_2) = (m_1 + m_2) v_{final}$$

Important Notes:

- Momentum is always conserved in isolated systems.
- Kinetic energy is NOT conserved in inelastic collisions.
- Internal energy and heat increase after sticking collisions.

Applications:

- Car crash analysis
- Gun recoil
- Rocket propulsion
- Sports: catching a ball, tackling

Advanced Mechanics – Detailed Theory & Concept Notes

1. Rolling Motion – Solid Sphere on Incline

Rolling motion is a combination of translation (movement of center of mass) and rotation about the center. For rolling without slipping, the point of contact with the ground is momentarily at rest. This creates a relation between linear velocity (v) and angular velocity (ω):

$$v = \omega r$$

Energy Distribution in Rolling:

When a body rolls, its total kinetic energy is the sum of translational KE and rotational KE. Thus a rolling object always has more energy than an object sliding at the same speed. This affects acceleration and final velocity.

Conservation of Mechanical Energy:

Initial Energy = Potential Energy (mgh)

Final Energy = Translational KE + Rotational KE

$$mgh = (1/2)mv^2 + (1/2)I\omega^2$$

Using $I(\text{sphere}) = (2/5)mr^2$ and $\omega = v/r$, solve to get:

$$v = \sqrt{(10/7)gh}$$

2. Gravitational Potential Energy & Gravitational Field Strength

Concept of Gravitational Potential Energy (GPE):

GPE represents the work done against gravity to bring a mass from a reference point to a given position. Near Earth's surface, the gravitational force is approximately constant. Hence GPE increases linearly with height.

General formula from universal gravitation:

$$U(r) = -GMm / r$$

Near Surface Approximation ($h \ll R$):

For small heights, the change in gravitational potential energy becomes proportional to height. This is because gravitational field strength does not vary significantly when h is small compared to Earth's radius.

$$\Delta U \approx mgh$$

Gravitational Field Strength (g):

Gravitational field strength is defined as the force experienced per unit mass. It represents how strong gravity is at a point in space.

$$g(r) = GM / r^2$$

3. Moment of Inertia (MOI) – Uniform Rod About One End

Moment of inertia quantifies rotational inertia: the resistance offered by a body to a change in its rotational motion. It depends on mass distribution relative to the axis of rotation. Greater the distribution of mass away from axis → higher MOI.

Derivation using integration:

$$I = \int x^2 dm = \int x^2 (M/L) dx$$

$$I = (1/3)ML^2$$

This higher value (compared to axis at center) is because mass is farther from the axis.

4. Geostationary Satellites & Orbital Velocity

Why Geostationary?

A geostationary satellite appears fixed relative to Earth. This is essential for communication, broadcasting, GPS correction signals etc. To remain fixed over a point, its angular velocity must match Earth's rotation.

Conditions:

- Satellite must orbit above Earth's equator.
- Must revolve west-to-east (same direction as Earth).
- Must have orbital period $T = 24$ hours.

Orbital Velocity Derivation:

Centripetal force = Gravitational force:

$$GMm / r^2 = mv^2 / r$$

$$v = \sqrt{GM / r}$$

This fundamental expression determines the speed required to maintain stable circular orbit.

5. Rotational Kinetic Energy – Solid Disk

Rotational kinetic energy arises when a rigid body spins about an axis. It complements translational KE and depends on the moment of inertia and angular velocity. Objects with larger MOI require more energy to rotate.

$$K = (1/2)I\omega^2$$

$$\text{For a uniform disk: } I = (1/2)mR^2$$

6. Centre of Mass – Two Body System

The centre of mass (CM) is the point where the entire mass of the system can be considered concentrated for analyzing translation. It represents the average position of all mass weighted by their magnitudes.

Formula:

$$R_{cm} = (m_1 r_1 + m_2 r_2) / (m_1 + m_2)$$

Example & Insight:

If one object is heavier, the CM lies closer to it. CM plays a key role in collisions, rotation, stability and planetary motion.

Fluid Mechanics, Thermodynamics & Kinetic Theory – Theory Notes

Bernoulli's Theorem

Bernoulli's theorem is derived from conservation of mechanical energy in an ideal fluid. It states that pressure energy, kinetic energy and potential energy per unit volume remain constant along a streamline.

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

Continuity Equation

The continuity equation represents conservation of mass for incompressible flow. The mass flow rate remains constant across any cross-section of a stream.

$$A_1 v_1 = A_2 v_2$$

Kelvin–Planck Statement

The Kelvin–Planck statement of the second law says that no heat engine can convert all absorbed heat into work. Some heat must always be rejected to a sink, making perfect efficiency impossible.

Buoyancy & Archimedes' Principle

A submerged body experiences an upward buoyant force equal to the weight of fluid displaced. This principle is used in ships, hydrometers, and floating objects.

$$F_b = \rho g V$$

Kinetic Theory of Gases

Gas pressure arises from continuous molecular collisions with the container walls. Using momentum transfer, the kinetic theory equation is obtained.

$$PV = \frac{1}{3} Nm \overline{v^2}$$

Pascal's Law

Pascal's law states that pressure applied to an enclosed fluid is transmitted equally in all directions. This enables hydraulic machines, brakes and lifts.

$$P = F / A$$

Pressure at Depth

Pressure in a liquid increases with depth due to the weight of liquid above the point of measurement.

$$P = \rho g h$$

Laws of Thermodynamics

First Law: Heat supplied to a system equals the sum of change in internal energy and work done.

$$\Delta Q = \Delta U + \Delta W$$

Second Law: Heat flows naturally from hot to cold. No cyclic device is perfectly efficient; entropy increases.

Waves, Optics & SHM – Detailed Rich Theory Notes

Standing Waves on a Stretched String

Standing waves are formed when two identical waves travelling in opposite directions superpose. On a stretched string with fixed ends, only certain wavelengths can persist—these correspond to the natural modes of vibration, or harmonics. Nodes (zero displacement) and antinodes (maximum displacement) form at fixed points, creating stationary wave patterns.

Wave speed on string:

$$v = \sqrt{T/\mu}$$

Here T is the tension applied to the string and μ is the linear mass density. This speed determines the allowed frequencies for standing waves.

Harmonic frequencies:

$$f_n = n(v / 2L)$$

Each harmonic corresponds to a unique pattern: higher harmonics have more loops and higher frequencies. These concepts apply to musical instruments like guitars, violins and pianos where rich sound results from multiple harmonics.

Single-Slit Diffraction

Diffraction refers to the bending and spreading of waves when they encounter an obstacle or slit. When light passes through a narrow slit comparable to its wavelength, the outgoing wavefront spreads widely instead of traveling straight. This produces a characteristic diffraction pattern on a screen.

Condition for minima:

$$a \sin\theta = m\lambda$$

The intensity distribution consists of a central bright maximum—twice as wide as the secondary maxima—followed by alternating minima and maxima. The diffraction pattern forms an envelope, with intensity gradually decreasing away from the center. This explains why optical instruments have resolution limits (diffraction limit).

Convex Lens – Image Formation

Convex lenses converge incoming parallel rays to a focal point. The formation of images depends on the position of the object relative to the focal points. A convex lens can form real, inverted images or virtual, upright images depending on distance.

Lens formula:

$$1/f = 1/v + 1/u$$

This relation is derived from ray geometry and remains fundamental in optics. Applications include cameras, microscopes, spectacles and telescopes.

Simple Harmonic Motion (SHM)

SHM is the simplest form of oscillatory motion where the restoring force is directly proportional to displacement and always directed towards equilibrium. Many physical systems—springs, pendulums, molecules—exhibit SHM for small oscillations.

Restoring force and DE:

$$F = -kx$$

$$m \, d^2x/dt^2 + kx = 0$$

The above differential equation has sinusoidal solutions, indicating periodic oscillations. The natural frequency is determined by $\omega = \sqrt{k/m}$. SHM forms the foundation of wave motion, alternating current theory, and resonance phenomena.

Combination of Lenses

When two lenses are placed in contact, their combined optical power equals the sum of individual powers. This simplifies the design of optical instruments by achieving desired focal lengths using multiple thin lenses.

$$1/F = 1/f_1 + 1/f_2$$

Doppler Effect

The Doppler effect is the apparent change in frequency of waves when there is relative motion between source and observer. If the source approaches, the wavefronts compress and frequency increases; if it moves away, frequency decreases. This phenomenon applies to sound, light and electromagnetic radiation.

General formula for apparent frequency:

$$f' = (v \pm v_o) / (v \mp v_s) \times f$$

Applications include radar guns, ultrasound imaging, astronomy (redshift and blueshift), weather radar and motion sensing devices.

Interference of Light

Interference occurs when two coherent light waves superpose to produce a pattern of bright and dark fringes. Constructive interference happens when the waves arrive in phase; destructive interference occurs when they are out of phase.

Bright fringes:

$$\Delta x = m\lambda$$

Dark fringes:

$$\Delta x = (m + 1/2)\lambda$$

Interference demonstrates the wave nature of light and is fundamental to technologies like thin-film coatings, interferometers, and optical metrology.

Concave Mirror – Image Formation

A concave mirror converges parallel rays to a focal point. Image formation depends on object distance: real inverted images form when the object lies beyond the focal point, and virtual enlarged images form when the object lies between mirror and focal point.

Mirror formula:

$$1/f = 1/v + 1/u$$